

Comp9417 Week4

Classification

Bingchen Xie

Disclaimer:

These slides are my own supplementary resource and are not official course materials. Any mistakes are my own.

Please refer to the solutions posted on **Moodle** as the official source.

Question 1

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

Q1a Analyze

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

(a) Calculate the probability of testing positive.

T: test positive

$$P(D) = ?$$

D: has the disease

$$P(T | D) = ?$$

$$P(T | \sim D) = ?$$

The probability of testing positive: $P(?)$

Q1a Analyze

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

(a) Calculate the probability of testing positive.

T: test positive

$$P(D) = 0.01$$

D: has the disease

$$P(T | D) = 0.95$$

$$P(T | \sim D) = 0.05$$

The probability of testing positive: $P(T)$

Q1a Analyze

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

(a) Calculate the probability of testing positive.

T: test positive

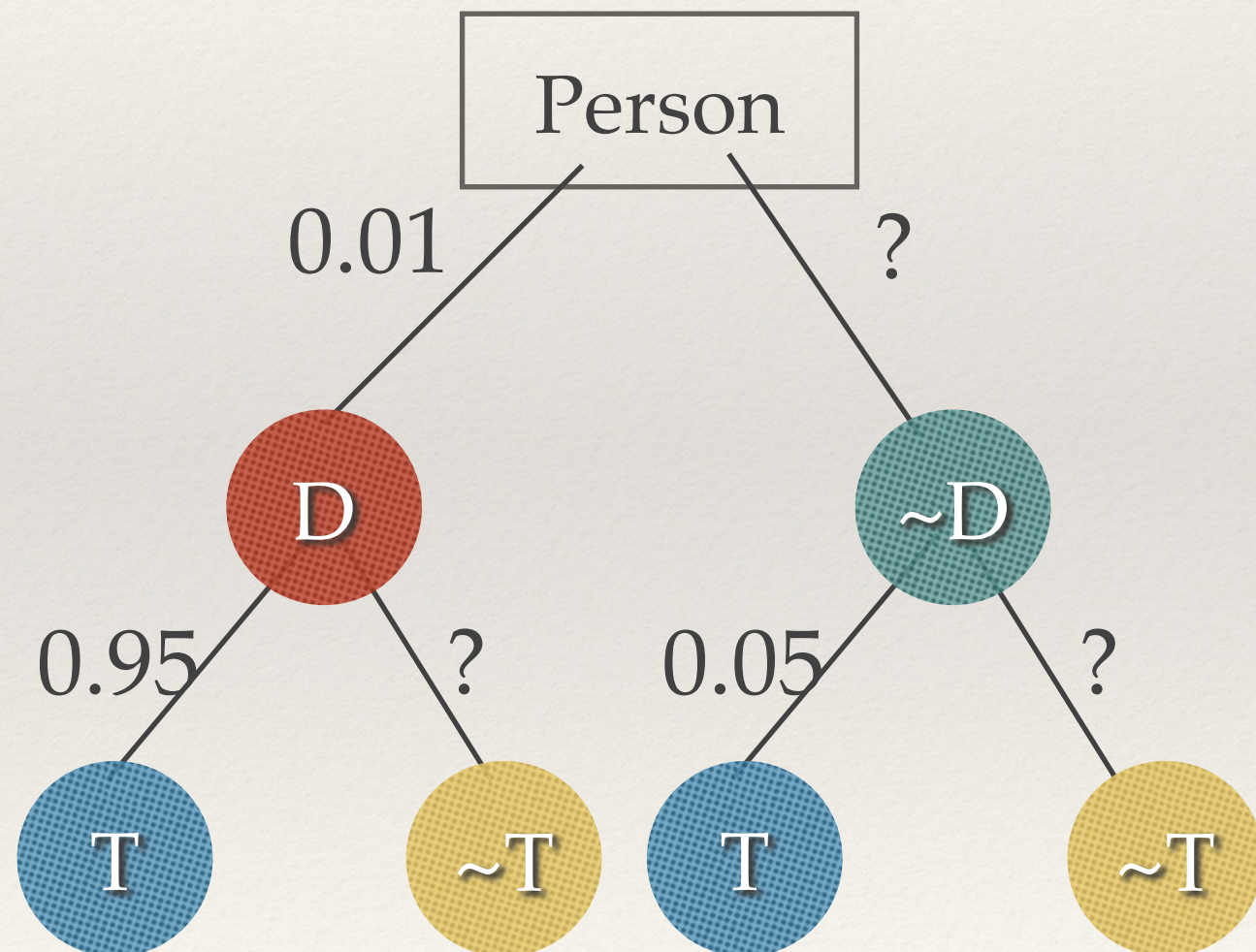
D: has the disease

$$P(D) = 0.01$$

$$P(T | D) = 0.95$$

$$P(T | \sim D) = 0.05$$

$$P(T) = ?$$



Q1a Analyze

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

(a) Calculate the probability of testing positive.

T: test positive

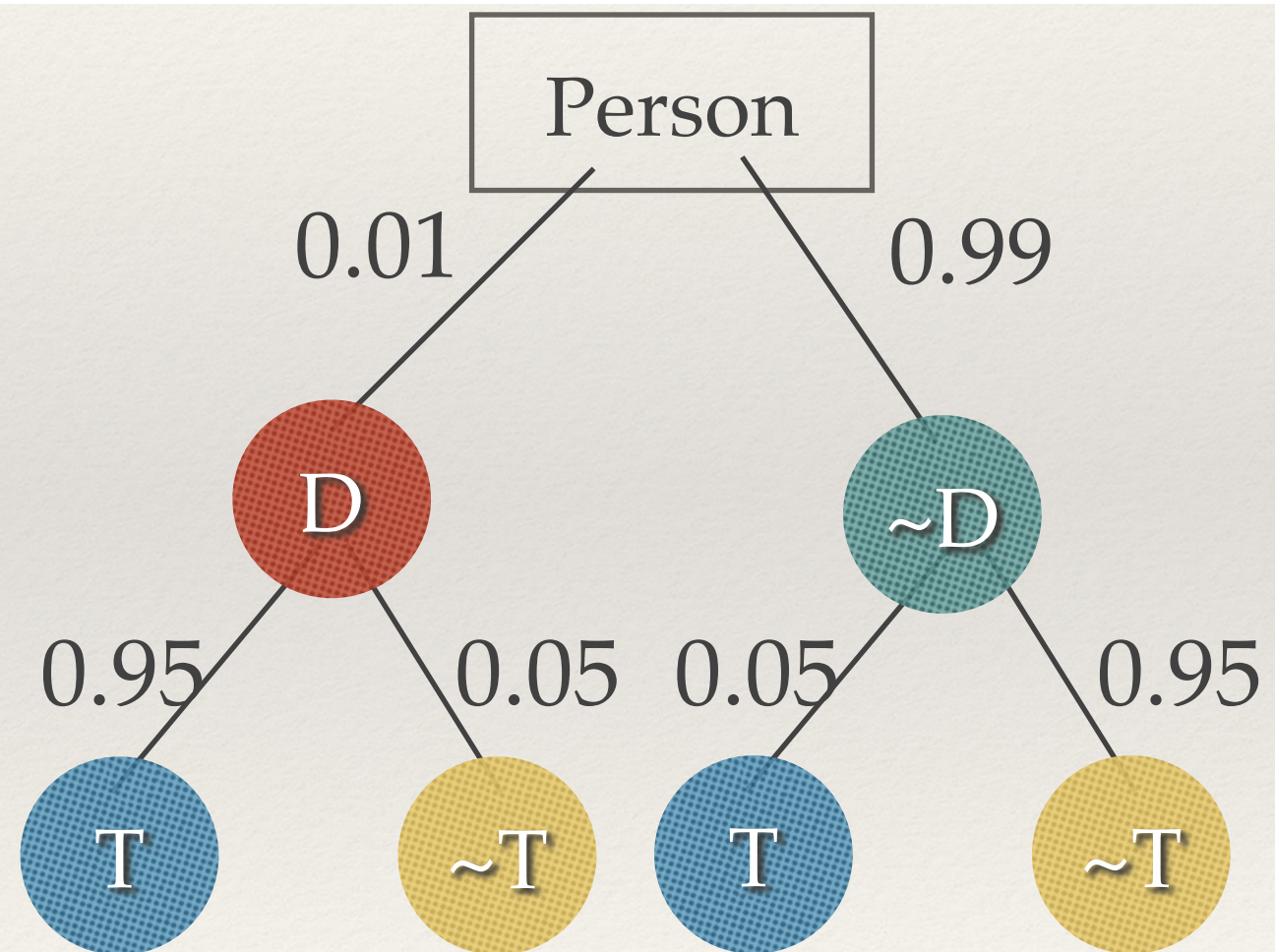
D: has the disease

$$P(D) = 0.01$$

$$P(T | D) = 0.95$$

$$P(T | \sim D) = 0.05$$

$$P(T) = ?$$



$$P(T \cap D) = P(T | D) * P(D) = 0.95 * 0.01 = 0.0095$$

Q1a Analyze

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

(a) Calculate the probability of testing positive.

T: test positive

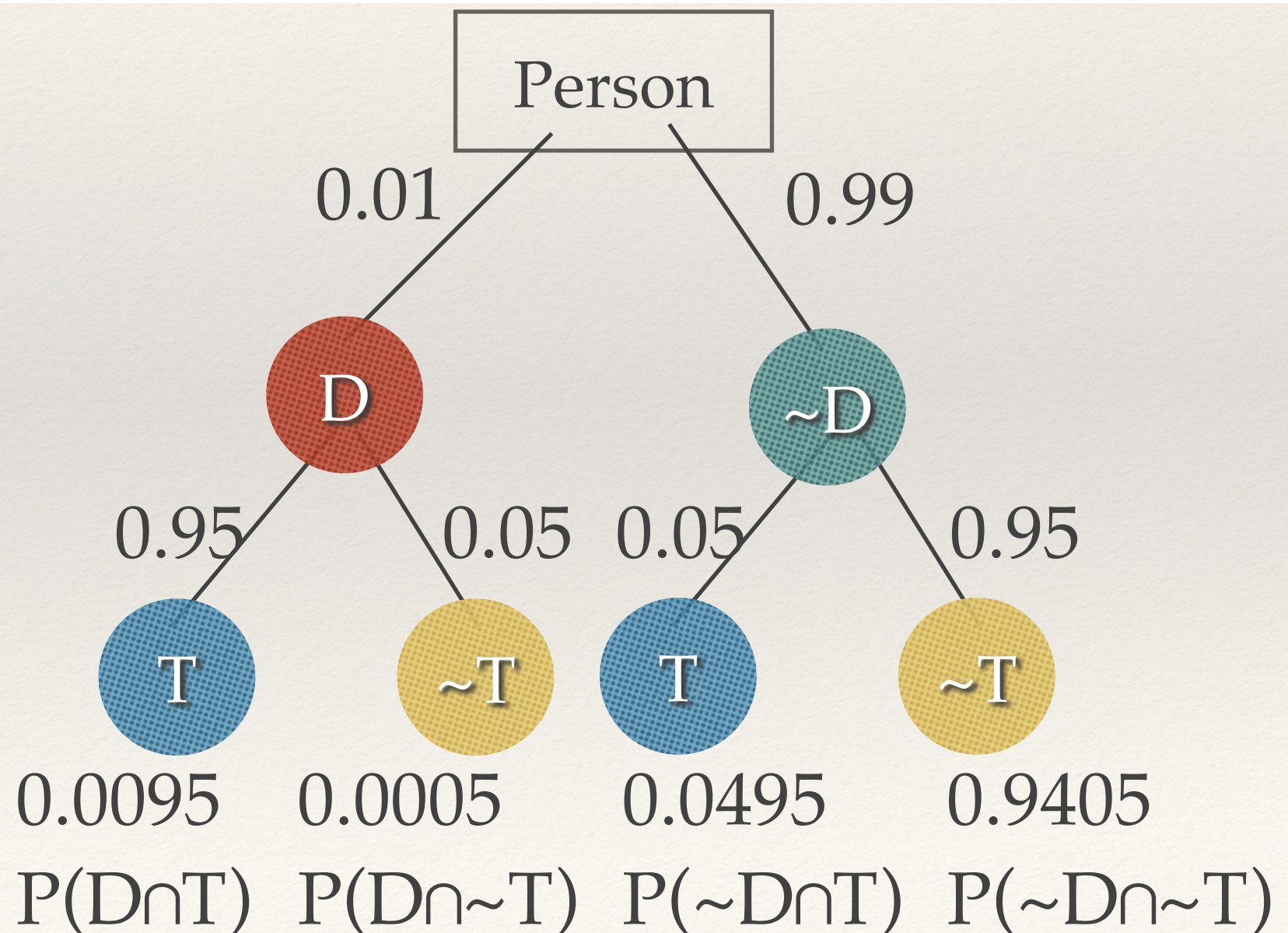
D: has the disease

$$P(D) = 0.01$$

$$P(T | D) = 0.95$$

$$P(T | \sim D) = 0.05$$

$$P(T) = ?$$



Q1a Calculation

Question 1 (Bayes Rule)

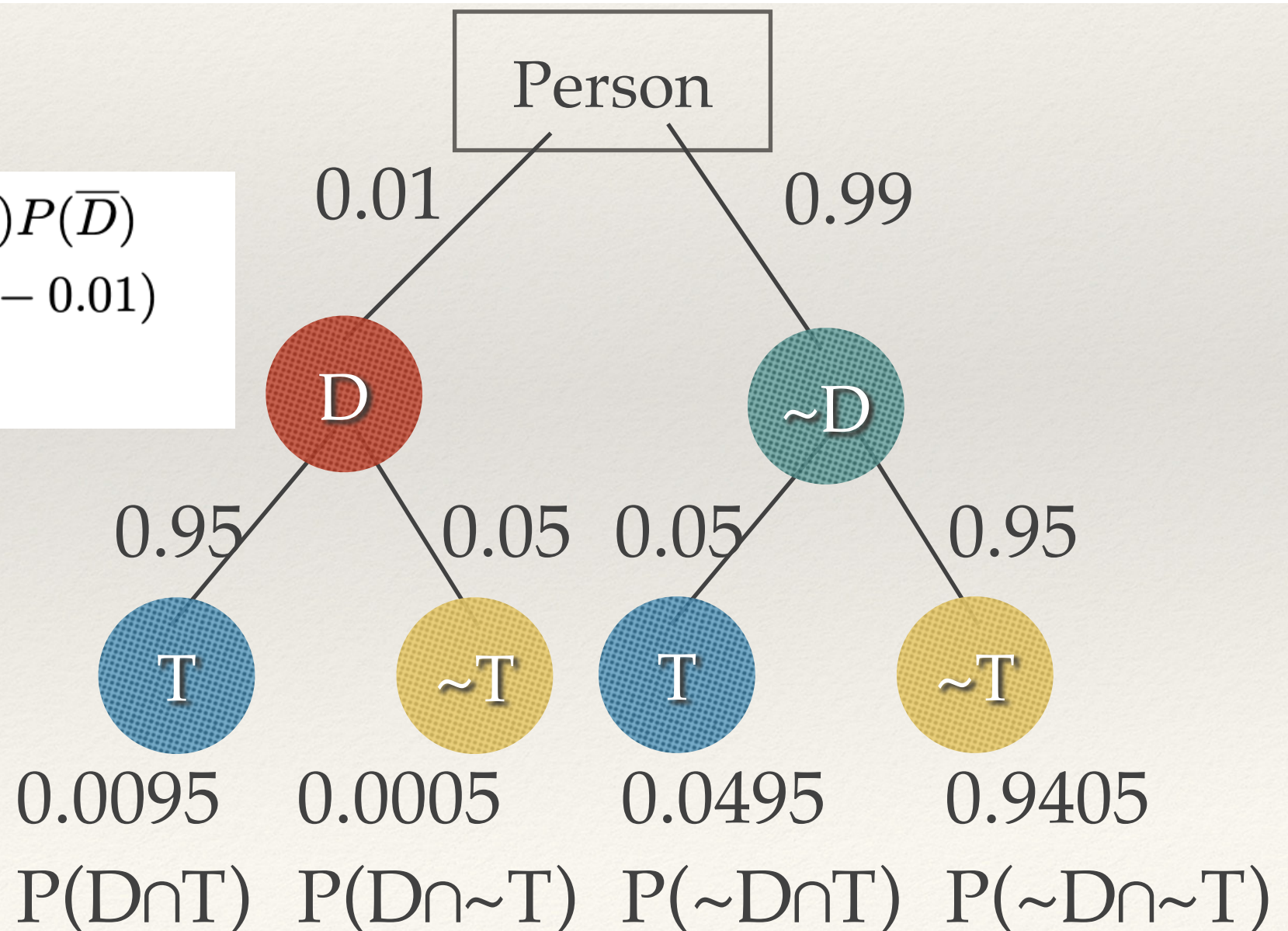
Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

(a) Calculate the probability of testing positive.

$$P(T) = ?$$

$$\begin{aligned} P(T) &= P(T|D)P(D) + P(T|\bar{D})P(\bar{D}) \\ &= 0.95 \times 0.01 + 0.05 \times (1 - 0.01) \\ &= 0.059. \end{aligned}$$

$$P(T) = 0.059$$



Q1b Analyze

(b) Calculate the probability of being infected with the disease, given that the test is positive.

$$P(? | ?)$$

T: test positive

D: has the disease

Q1b Analyze

(b) Calculate the probability of being infected with the disease, given that the test is positive.

$$P(D | T)$$

T: test positive

D: has the disease

From Q1a:

$$P(D) = 0.01$$

$$P(T | D) = 0.95$$

$$P(T | \sim D) = 0.05$$

Q1b Analyze

(b) Calculate the probability of being infected with the disease, given that the test is positive.

$$P(D | T) = ?$$

Chain Rule

$$\begin{aligned} P(D \cap T) &= P(D | T) * P(T) \\ P(D \cap T) &= P(T | D) * P(D) \end{aligned}$$

$$P(D | T) * P(T) = P(T | D) * P(D)$$



Bayes Rule

$$P(D | T) = P(T | D) * P(D) / P(T)$$

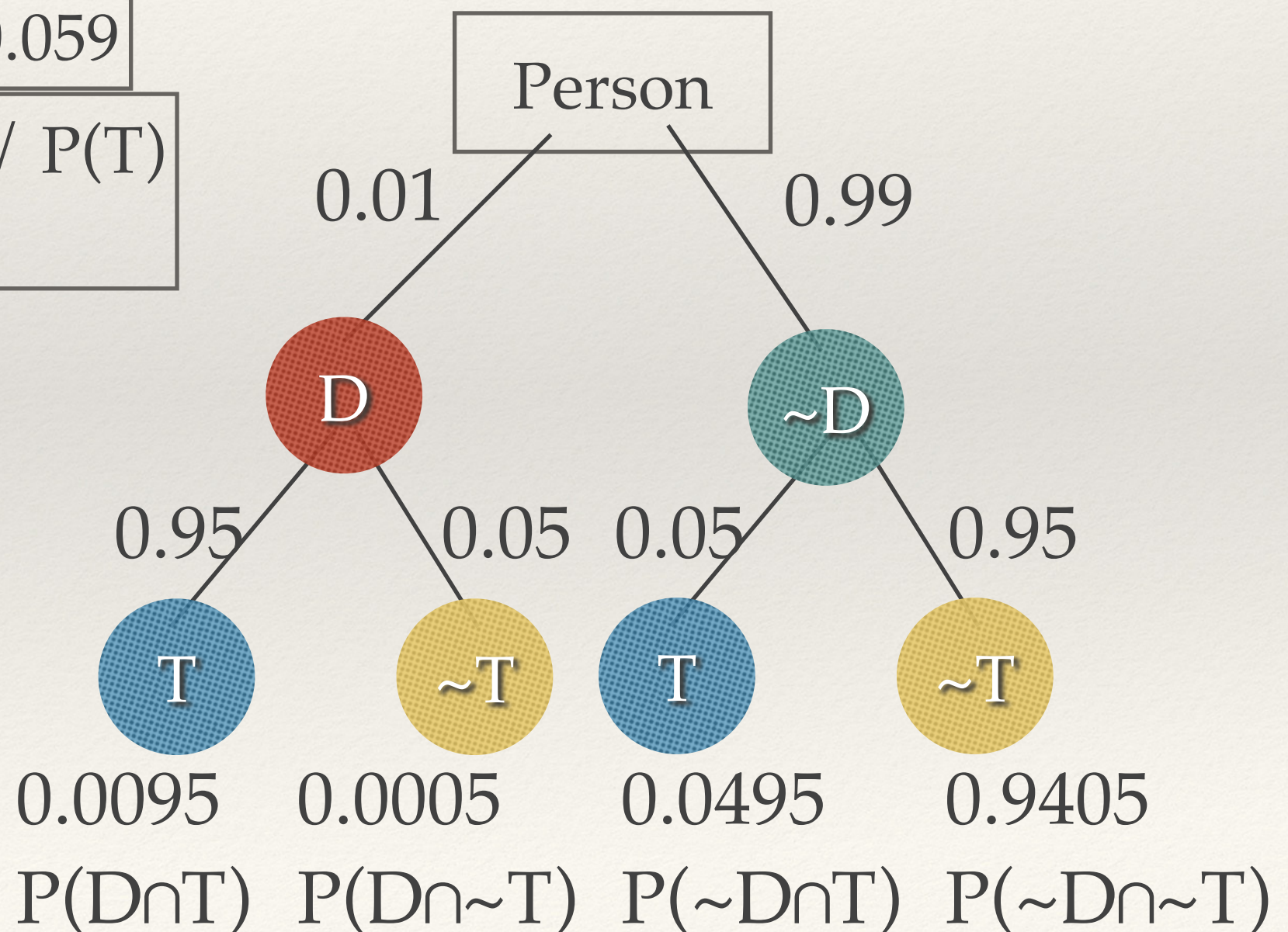
Q1b Analyze

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

$$P(T) = 0.0095 + 0.0495 = 0.059$$

$$\begin{aligned} P(D | T) &= P(T | D) * P(D) / P(T) \\ &= P(D \cap T) / P(T) \end{aligned}$$



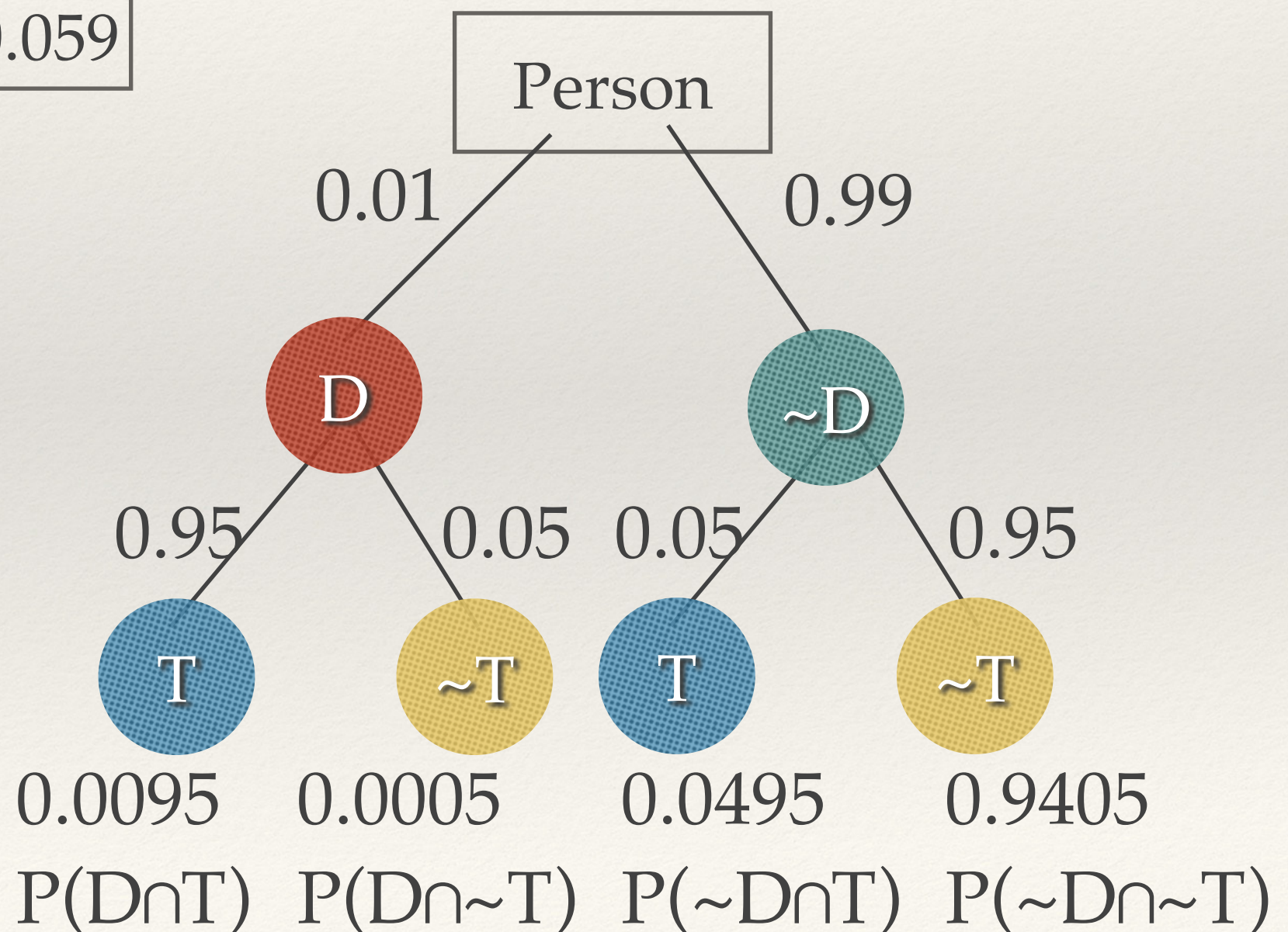
Q1b Calculation

Question 1 (Bayes Rule)

Assume that the probability of a certain disease is 0.01. The probability of testing positive given that a person is infected with the disease is 0.95, and the probability of testing positive given that the person is not infected with the disease is 0.05.

$$P(T) = 0.0095 + 0.0495 = 0.059$$

$$\begin{aligned} P(D|T) &= \frac{P(D \cap T)}{P(T)} \\ &= \frac{P(T|D)P(D)}{P(T)} \\ &= \frac{0.95 \times 0.01}{0.059} \\ &= 0.16, \end{aligned}$$



Q1c Analyze

(c) Now assume that you test the individual a second time, and the test comes back positive (so two tests, two positives). Assume that conditional on having the disease, the outcomes of the two tests are independent, what is the probability that the individual has the disease? (note, conditional independence in this case means that $P(TT|D) = P(T|D)P(T|D)$, and not $P(TT) = P(T)P(T)$.) You may also assume that the test outcomes are conditionally independent given not having the disease.

$P(? | ?)$

T: test positive

D: has the disease

Q1c Analyze

(c) Now assume that you test the individual a second time, and the test comes back positive (so two tests, two positives). Assume that conditional on having the disease, the outcomes of the two tests are independent, what is the probability that the individual has the disease? (note, conditional independence in this case means that $P(TT|D) = P(T|D)P(T|D)$, and not $P(TT) = P(T)P(T)$.) You may also assume that the test outcomes are conditionally independent given not having the disease.

$P(D | TT)$

T: test positive

D: has the disease

Q1c Analyze

(c) Now assume that you test the individual a second time, and the test comes back positive (so two tests, two positives). Assume that conditional on having the disease, the outcomes of the two tests are independent, what is the probability that the individual has the disease? (note, conditional independence in this case means that $P(TT|D) = P(T|D)P(T|D)$, and not $P(TT) = P(T)P(T)$.) You may also assume that the test outcomes are conditionally independent given not having the disease.

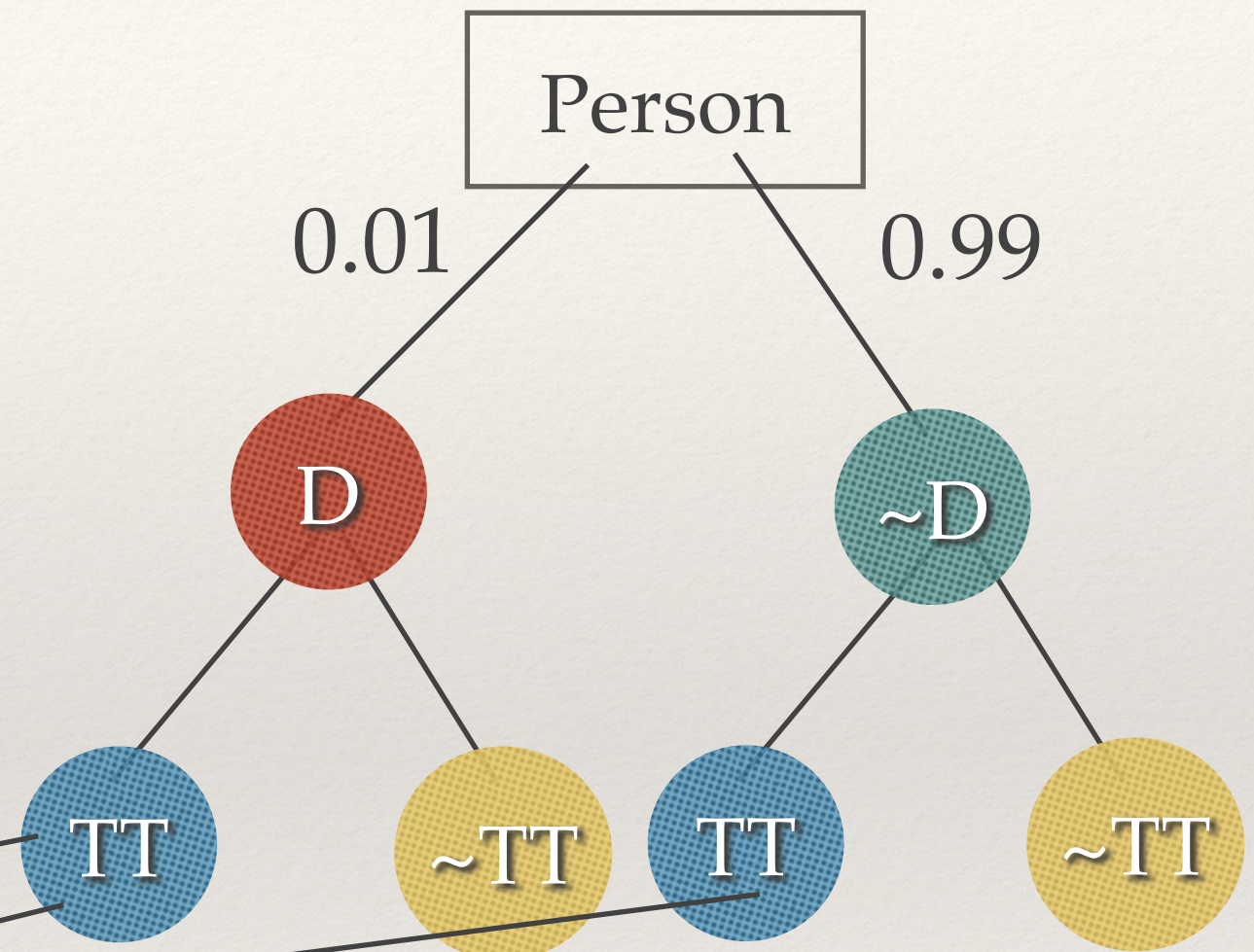
Bayes Rule

$$\begin{aligned} P(D | TT) &= P(TT | D) * P(D) / P(TT) \\ &= P(TT \cap D) / P(TT) \end{aligned}$$

Q1c Analyze

Bayes Rule

$$P(D | TT) = \\ = P(TT \cap D) / P(TT)$$



Q1c Calculation

(c) Now assume that you test the individual a second time, and the test comes back positive (so two tests, two positives). Assume that conditional on having the disease, the outcomes of the two tests are independent, what is the probability that the individual has the disease? (note, conditional independence in this case means that $P(TT|D) = P(T|D)P(T|D)$, and not $P(TT) = P(T)P(T)$.) You may also assume that the test outcomes are conditionally independent given not having the disease.

Bayes Rule: $P(D | TT) = P(TT \cap D) / P(TT)$

Conditional Independence:

$$P(TT | D) = P(T | D)P(T | D)$$
$$P(TT | \sim D) = P(T | \sim D)P(T | \sim D)$$

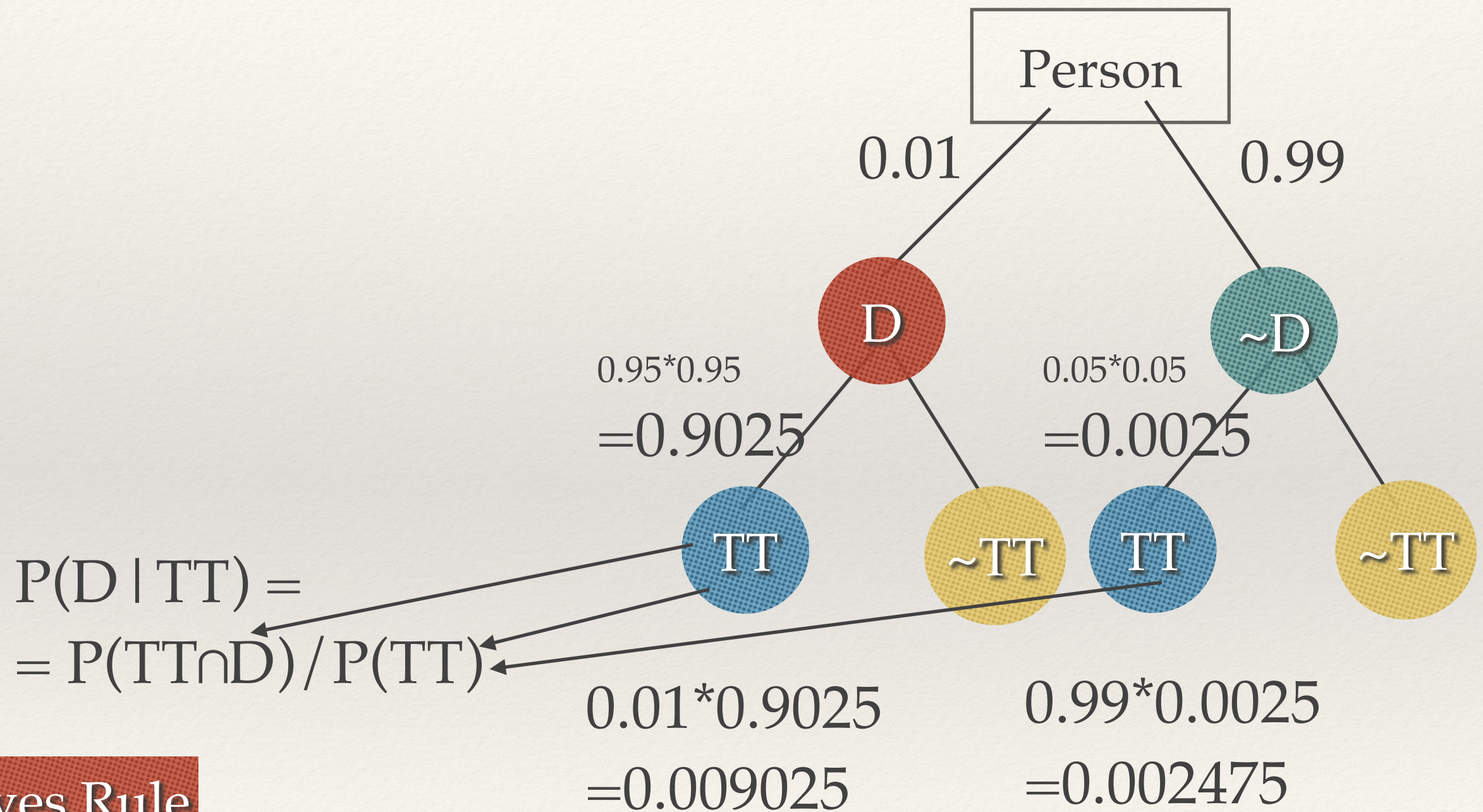
From Q1a:

$$P(D) = 0.01$$

$$P(T | D) = 0.95$$

$$P(T | \sim D) = 0.05$$

Q1c Analyze



Bayes Rule

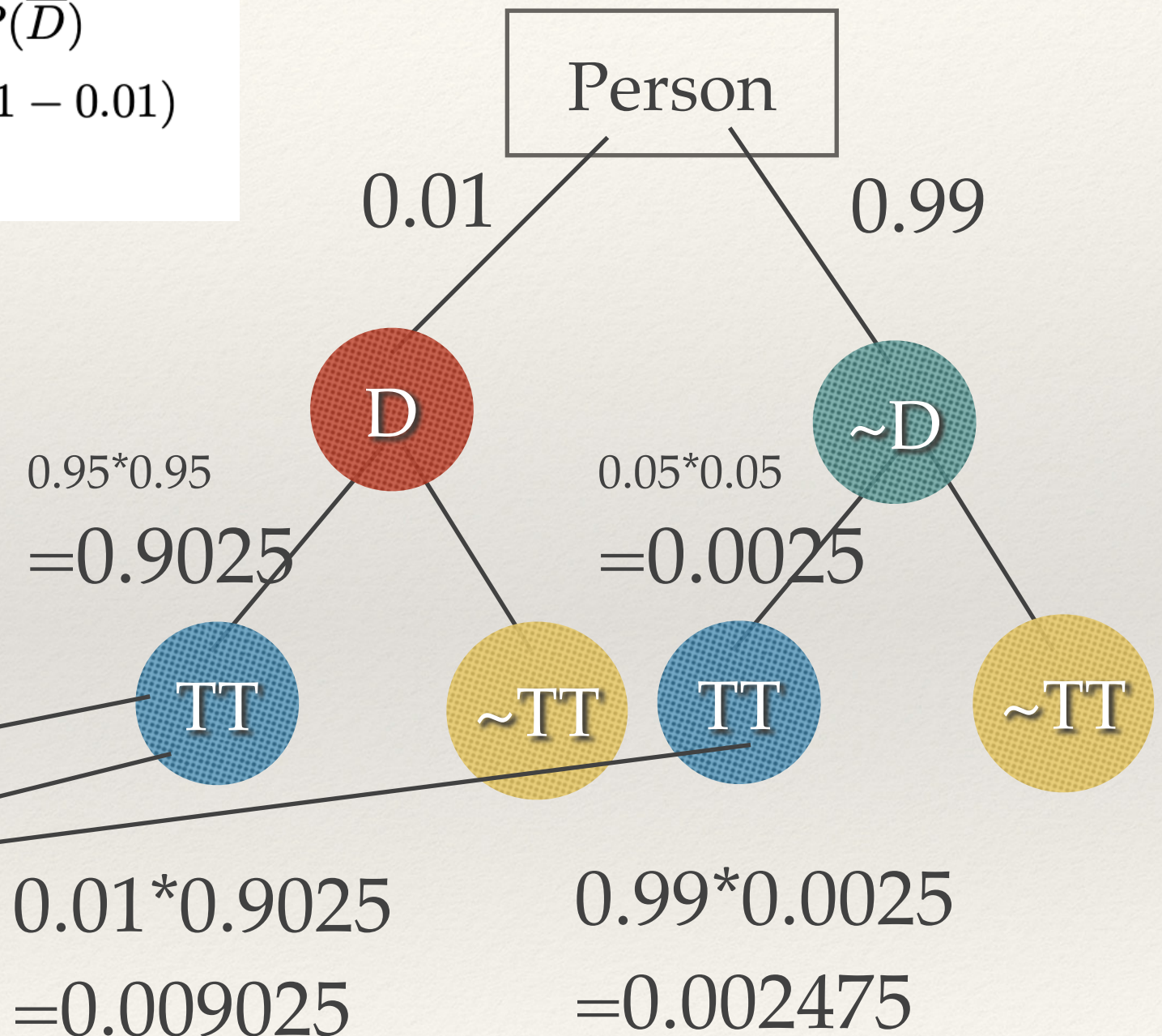
Q1c Calculation

$$\begin{aligned}P(TT) &= P(TT|D)P(D) + P(TT|\bar{D})P(\bar{D}) \\&= P(T|D)^2P(D) + P(T|\bar{D})^2P(\bar{D}) \\&= (0.95)^2 \times 0.01 + (0.05)^2 \times (1 - 0.01) \\&= 0.0115.\end{aligned}$$

$$\begin{aligned}P(D|TT) &= \frac{P(TT|D)P(D)}{P(TT)} \\&= \frac{(0.95)^2 \times 0.01}{0.0115} \\&= 0.7848.\end{aligned}$$

$$\begin{aligned}P(D | TT) &= \\&= P(TT \cap D) / P(TT)\end{aligned}$$

Bayes Rule



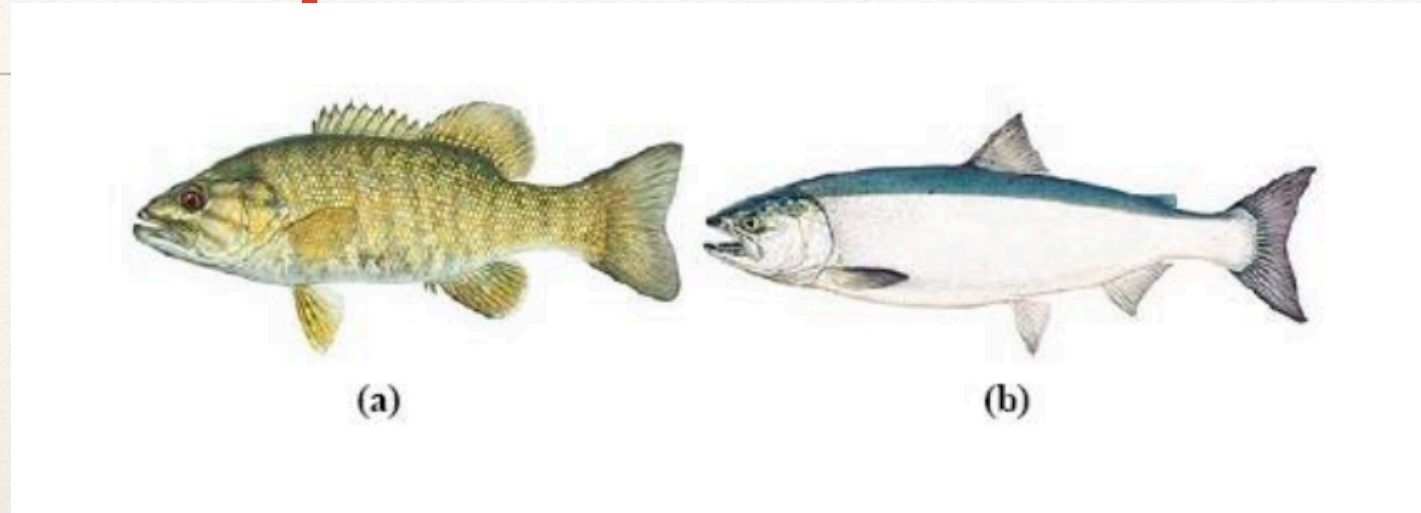
Question 2

Question 2 (Lecture Review)

In this question, we will review some important ideas from the lecture.

- (a) What is probabilistic classification? How does it differ from non-probabilistic classification methods?

Recap: classification



Classes:

$$C = \{c_1, \dots, c_K\}$$

$$C = \{\text{sea bass, salmon}\}, \quad K = 2$$

Dataset:

$$D = \{(\mathbf{x}_i, c_i) : i = 1, \dots, n\}$$

(Features, Label)

Features:

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{pmatrix}$$

x_{i1}

x_{i2}

x_{i3}

x_{i4}

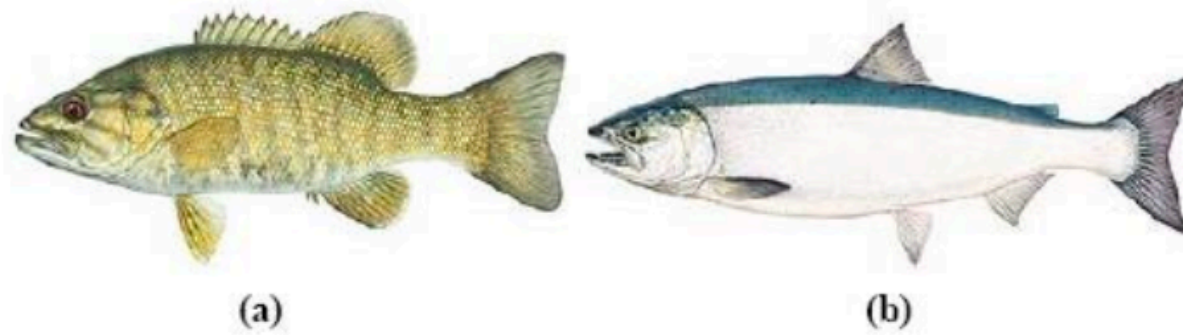
Width

Length

Weight

Lightness

Recap: classification



Dataset:

Features / input

Class / Label

Fish i	Width (cm)	Length (cm)	Weight (g)	Lightness	Class y_i
1	8.0	30	450	0.35	Sea bass
2	7.5	28	400	0.40	Sea bass
3	9.0	35	650	0.30	Sea bass
4	7.0	32	480	0.75	Salmon
5	8.5	38	720	0.80	Salmon
6	8.0	36	620	0.70	Salmon

Recap: classification

A new fish: $\mathbf{x}_\star$

Width (cm)	Length (cm)	Weight (g)	Lightness
8.2	31	490	0.38

Predict the class / label.

Q2a Analyze

Question 2 (Lecture Review)

In this question, we will review some important ideas from the lecture.

- (a) What is probabilistic classification? How does it differ from non-probabilistic classification methods?

$$p(c_k | \mathbf{x}_\star).$$

$$c_k$$

Probabilistic classification

Estimates class probabilities.

Example: $P(\text{salmon} | \mathbf{x}) = 0.8$, $P(\text{sea bass} | \mathbf{x}) = 0.2 \rightarrow \text{Salmon}$.

Provides class probabilities to express uncertainty.

Non-probabilistic classification

Predicts a class directly.

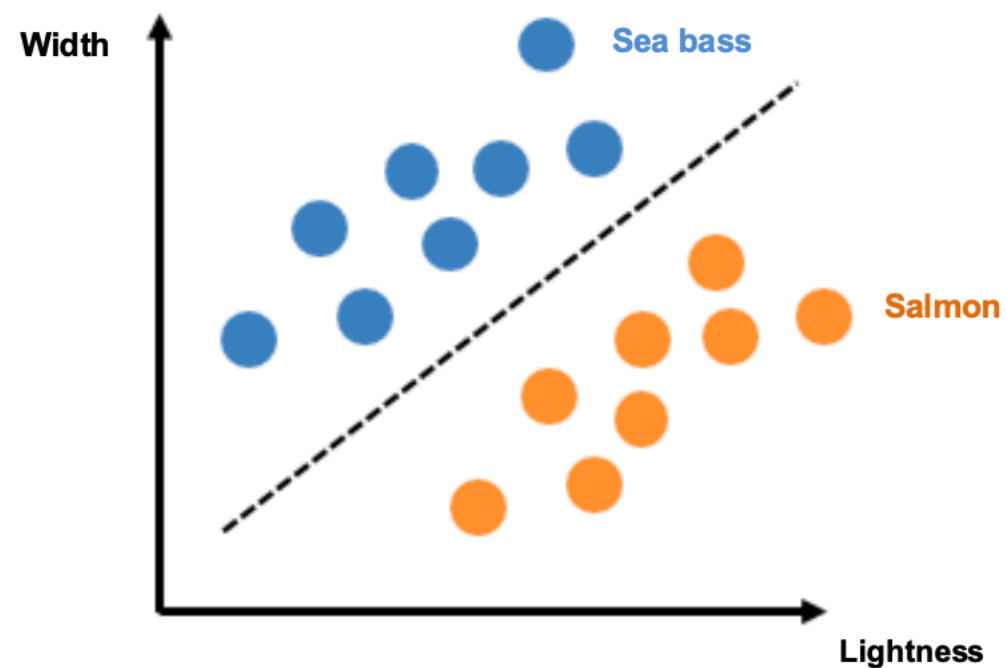
Example: most nearby fish are salmon $\rightarrow$ **Salmon**.

A class label alone does not express uncertainty.

Q2a Analyze

Classification

Example: Maybe we can find a line that separates the two classes.



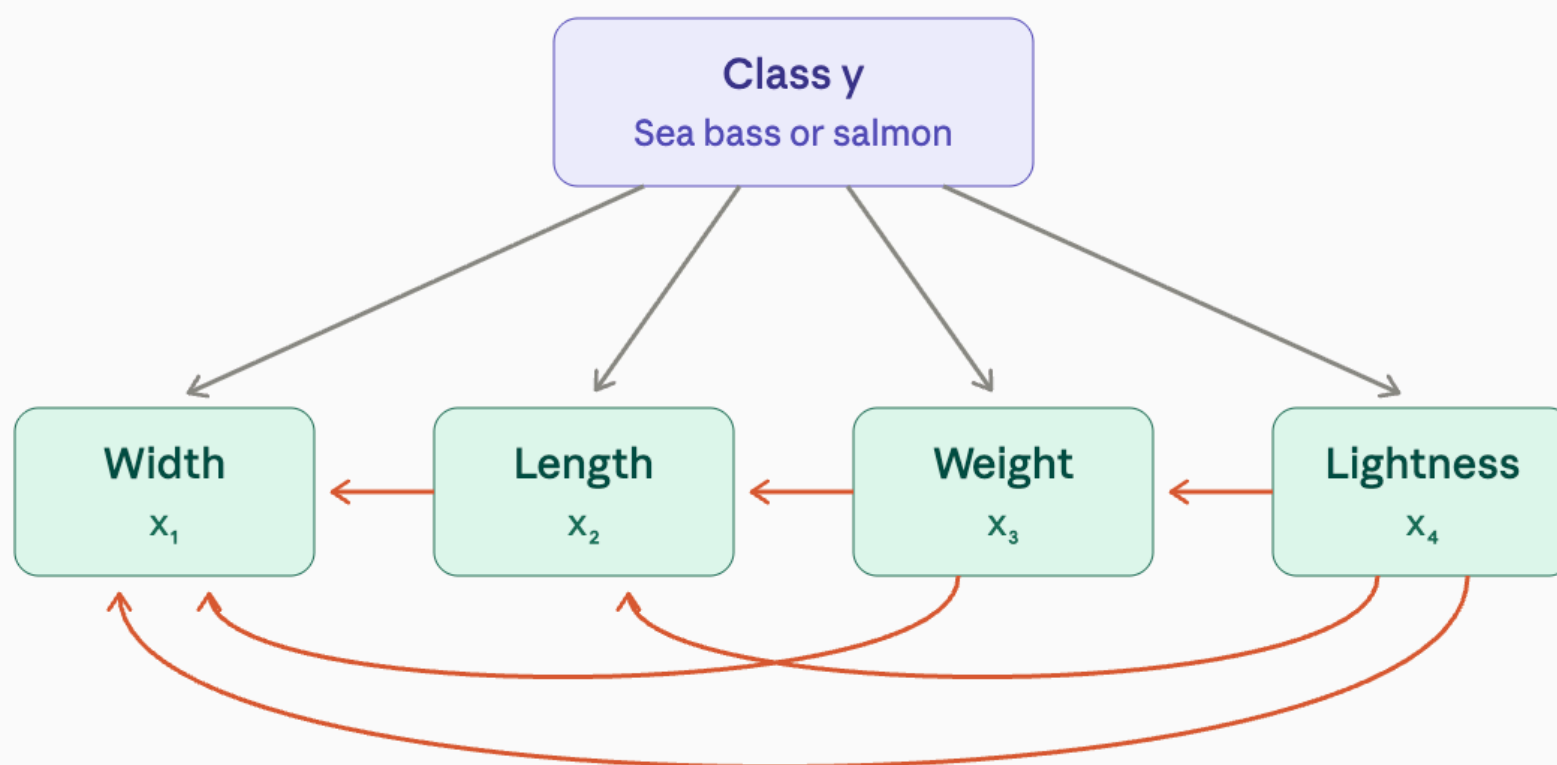
Q2b Analyze

(b) What is the Naive Bayes assumption and why do we need it?

Chain Rule

$$\begin{aligned} p(\mathbf{x} \mid c_k) p(c_k) &= p(x_1, x_2, x_3, x_4, c_k) \\ &= p(x_1 \mid x_2, x_3, x_4, c_k) p(x_2 \mid x_3, x_4, c_k) p(x_3 \mid x_4, c_k) p(x_4 \mid c_k) p(c_k). \end{aligned}$$

General case (no assumptions)



Q2b Analyze

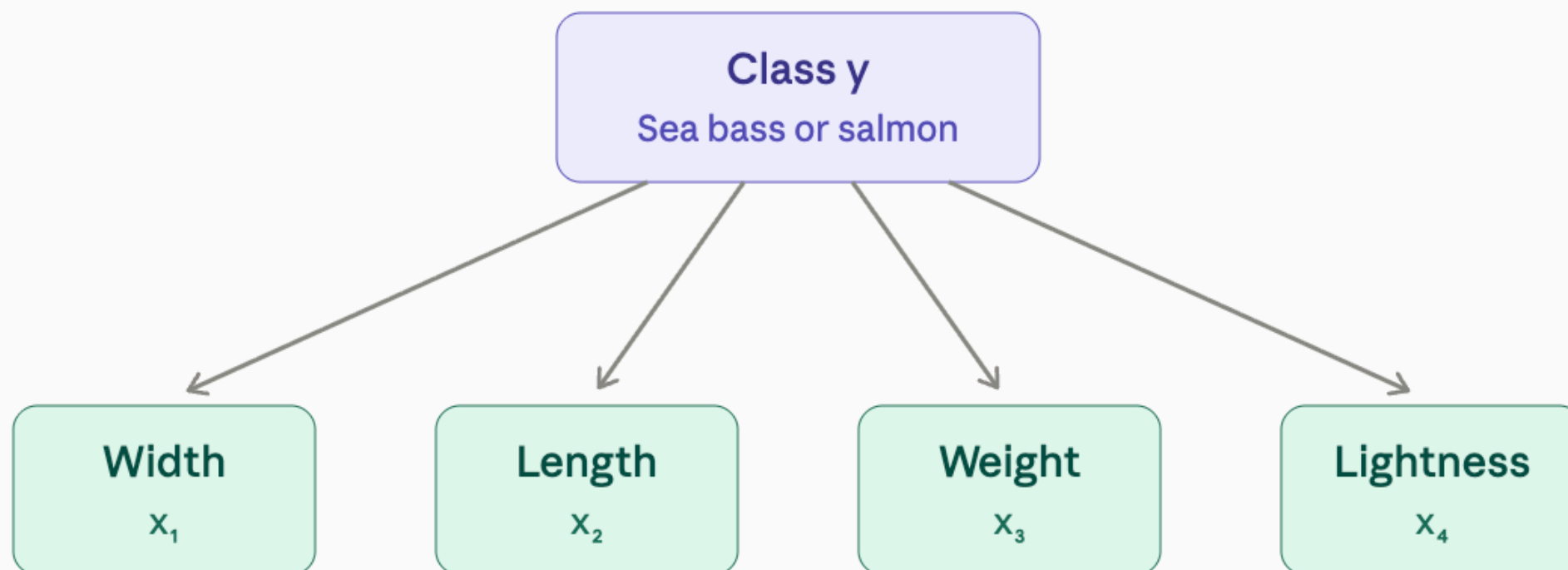
(b) What is the Naive Bayes assumption and why do we need it?

Conditional Independence:

$$x_i \perp x_j | c_k \quad \text{for all } j \neq i.$$

$$p(\mathbf{x} | c_k)p(c_k) = p(x_1 | c_k)p(x_2 | c_k)p(x_3 | c_k)p(x_4 | c_k)p(c_k).$$

Naive Bayes assumption



Recap: Discriminative Model

$$p(c_k | \mathbf{x}_\star).$$

Fish i	Width (cm)	Length (cm)	Weight (g)	Lightness	Class y_i
1	8.0	30	450	0.35	Sea bass
2	7.5	28	400	0.40	Sea bass
3	9.0	35	650	0.30	Sea bass
4	7.0	32	480	0.75	Salmon
5	8.5	38	720	0.80	Salmon
6	8.0	36	620	0.70	Salmon

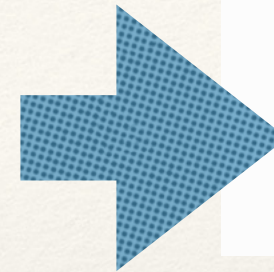
Logistic Regression

$$\mathbf{x} \xrightarrow{\text{Linear} + \text{sigmoid}} p(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}}.$$

Recap: Generative Model

Bayes Rule

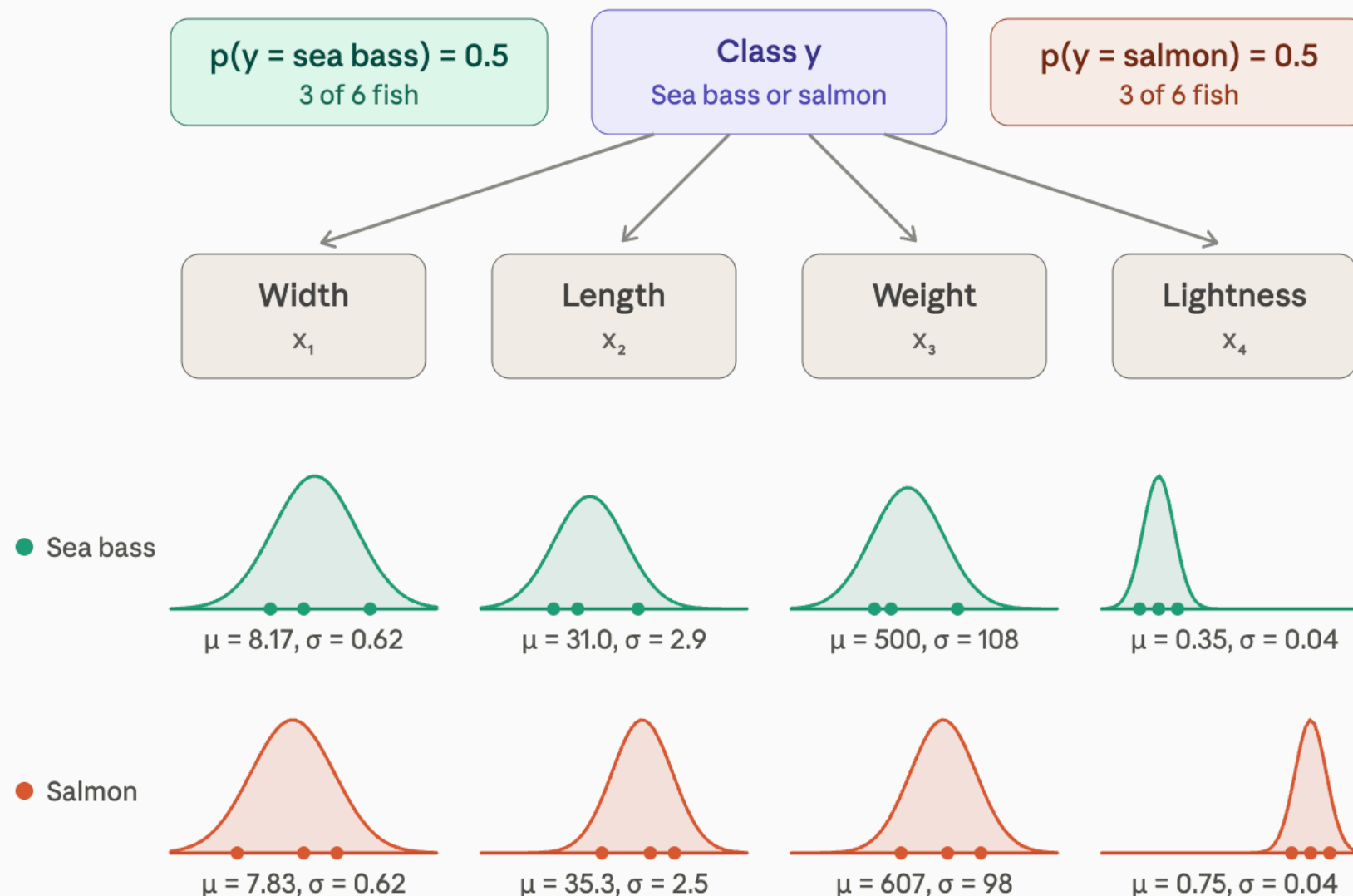
$$p(c_k | \mathbf{x}) = \frac{p(\mathbf{x} | c_k) p(c_k)}{p(\mathbf{x})}$$



$$p(c_k | \mathbf{x}) \propto p(\mathbf{x} | c_k) p(c_k).$$

From Q2a:

$$p(\mathbf{x} | c_k) p(c_k) = p(x_1 | c_k) p(x_2 | c_k) p(x_3 | c_k) p(x_4 | c_k) p(c_k).$$



Q2c

Bernoulli Naive Bayes

- (c) Consider the problem from lectures of classifying emails as **spam** or **ham**, with training data summarised below: Each row represents an email, and each email is a combination of words taken

e_1	b	d	e	b	b	d	e			
e_2	b	c	e	b	b	d	d	e	c	c
e_3	a	d	a	d	e	a	e	e		
e_4	b	a	d	b	e	d	a	b		
e_5	a	b	a	b	a	b	a	e	d	
e_6	a	c	a	c	a	c	a	e	d	
e_7	e	a	e	d	a	e	a			
e_8	d	e	d	e	d					

from the set $\{a, b, c, d, e\}$. We treat the words d, e as stop words - these are words that are not useful for classification purposes, for example, the word 'the' is too common to be useful for classifying documents as spam or ham. We therefore define our vocabulary as $V = \{a, b, c\}$. Note that in this case we have two classes, so $k = 2$, and we will assume a uniform prior, that is:

$$p(c_+) = p(c_-) = \frac{1}{2},$$

where $c_+ = \text{spam}$, $c_- = \text{ham}$. Review the multivariate Bernoulli Naive Bayes set-up and classify the test example: assume we get a new email that we want to classify: $e_\star = abbdebb$

e_1	b	d	e	b	b	d	e			
e_2	b	c	e	b	b	d	d	e	c	c
e_3	a	d	a	d	e	a	e	e		
e_4	b	a	d	b	e	d	a	b		
e_5	a	b	a	b	a	b	a	e	d	
e_6	a	c	a	c	a	c	a	e	d	
e_7	e	a	e	d	a	e	a			
e_8	d	e	d	e	d					

Q2c Analyze

Step1: Take a look at the dataset

	x_{ia}	x_{ib}	x_{ic}
e_1	0	1	0
e_2	0	1	1
e_3	1	0	0
e_4	1	1	0
e_5	1	1	0
e_6	1	0	1
e_7	1	0	0
e_8	0	0	0

Email	x_{ia}	x_{ib}	x_{ic}	Label y_i
e_1	0	1	0	spam (c_+)
e_2	0	1	1	spam (c_+)
e_3	1	0	0	spam (c_+)
e_4	1	1	0	spam (c_+)
e_5	1	1	0	ham (c_-)
e_6	1	0	1	ham (c_-)
e_7	1	0	0	ham (c_-)
e_8	0	0	0	ham (c_-)

$e_\star = abbdebb$

?

?

?

?

Q2c Analyze

Step1: Take a look at the dataset

	x_{ia}	x_{ib}	x_{ic}
e_1	0	1	0
e_2	0	1	1
e_3	1	0	0
e_4	1	1	0
e_5	1	1	0
e_6	1	0	1
e_7	1	0	0
e_8	0	0	0

Email	x_{ia}	x_{ib}	x_{ic}	Label y_i
e_1	0	1	0	spam (c_+)
e_2	0	1	1	spam (c_+)
e_3	1	0	0	spam (c_+)
e_4	1	1	0	spam (c_+)
e_5	1	1	0	ham (c_-)
e_6	1	0	1	ham (c_-)
e_7	1	0	0	ham (c_-)
e_8	0	0	0	ham (c_-)

$e_\star = abbdebb$

1

1

0

?

Q2c Analyze

Step2: Write down the probabilities

Just Count

Sapm

Ham

	x_{ia}	x_{ib}	x_{ic}
e_1	0	1	0
e_2	0	1	1
e_3	1	0	0
e_4	1	1	0
e_5	1	1	0
e_6	1	0	1
e_7	1	0	0
e_8	0	0	0

$$p_a^+ = \frac{2}{4}, \quad p_b^+ = \frac{3}{4}, \quad p_c^+ = \frac{1}{4},$$

$$p_a^- = \frac{3}{4}, \quad p_b^- = \frac{1}{4}, \quad p_c^- = \frac{1}{4}.$$

Q2c Analyze

Step2: Write down the probabilities

Spam

Ham

	x_{ia}	x_{ib}	x_{ic}
e_1	0	1	0
e_2	0	1	1
e_3	1	0	0
e_4	1	1	0
e_5	1	1	0
e_6	1	0	1
e_7	1	0	0
e_8	0	0	0

$$P_{a+} = P(X_a=1 \mid S+) = P(X_a=1 \mid \text{Spam})$$

$$p_a^+ = \frac{2}{4}, \quad p_b^+ = \frac{3}{4}, \quad p_c^+ = \frac{1}{4},$$

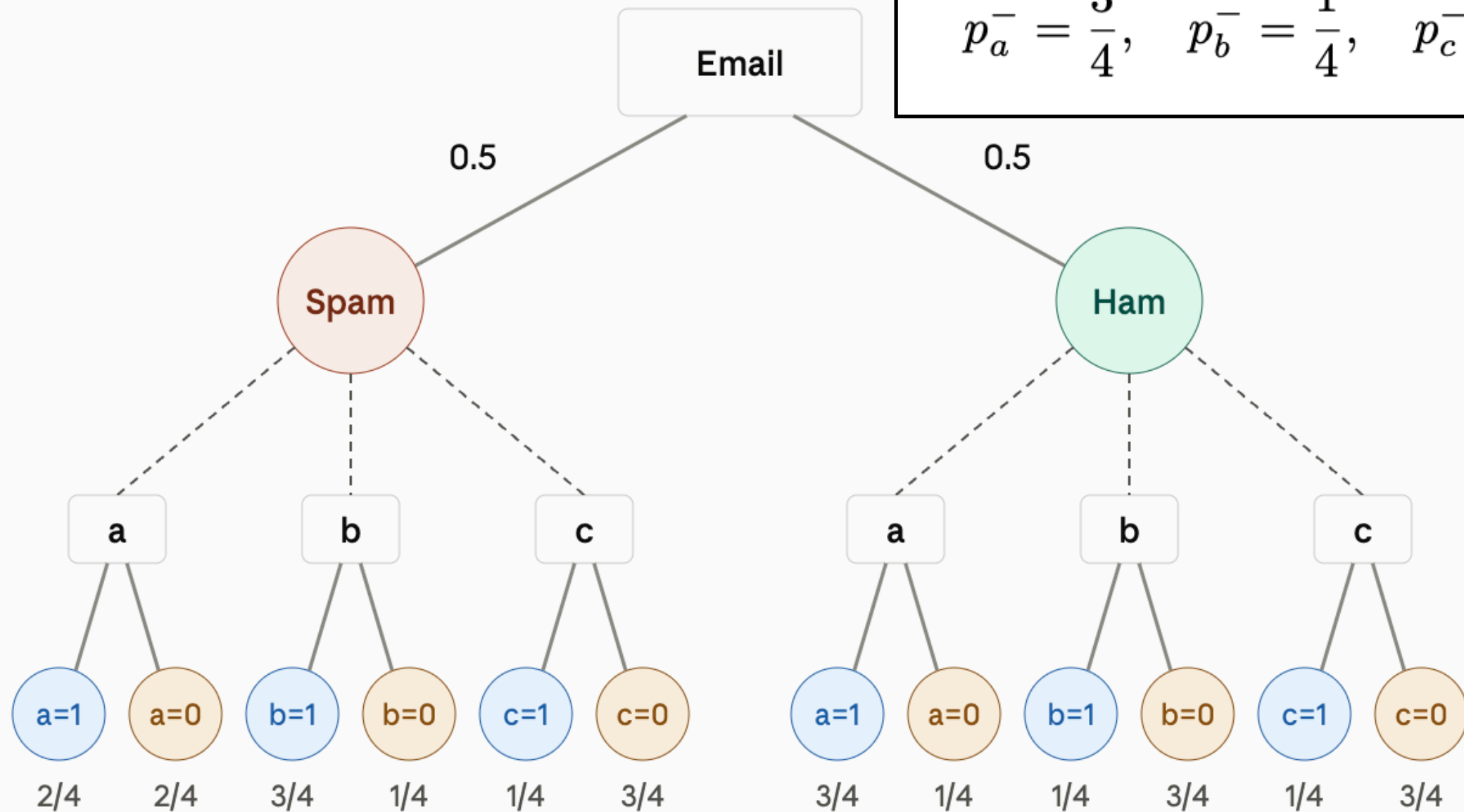
$$p_a^- = \frac{3}{4}, \quad p_b^- = \frac{1}{4}, \quad p_c^- = \frac{1}{4}.$$

$$P_{a-} = P(X_a=1 \mid S-) = P(X_a=1 \mid \sim \text{Spam})$$

Q2c Analyze

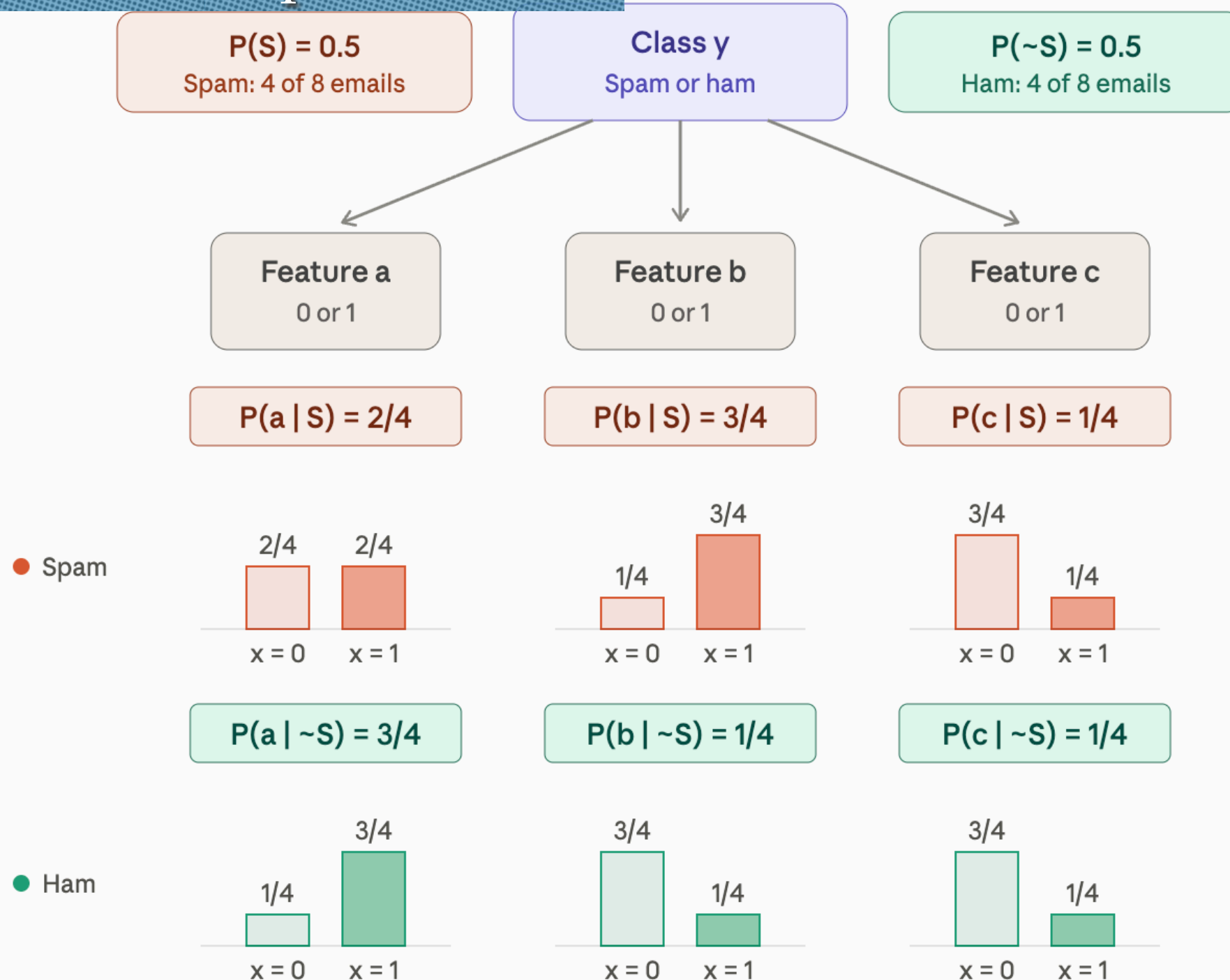
Step2: Write down the probabilities

$$\begin{aligned} p_a^+ &= \frac{2}{4}, & p_b^+ &= \frac{3}{4}, & p_c^+ &= \frac{1}{4}, \\ p_a^- &= \frac{3}{4}, & p_b^- &= \frac{1}{4}, & p_c^- &= \frac{1}{4}. \end{aligned}$$



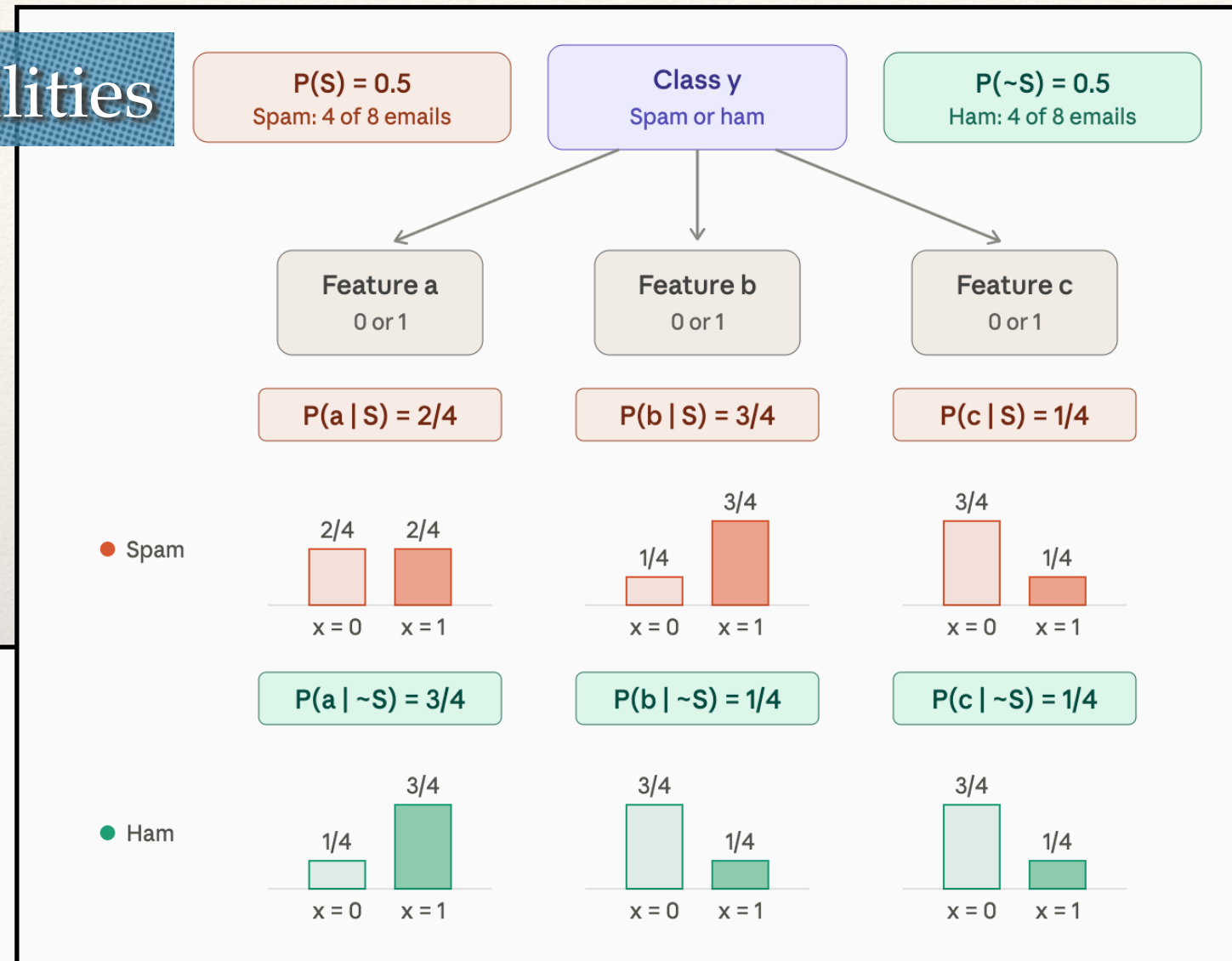
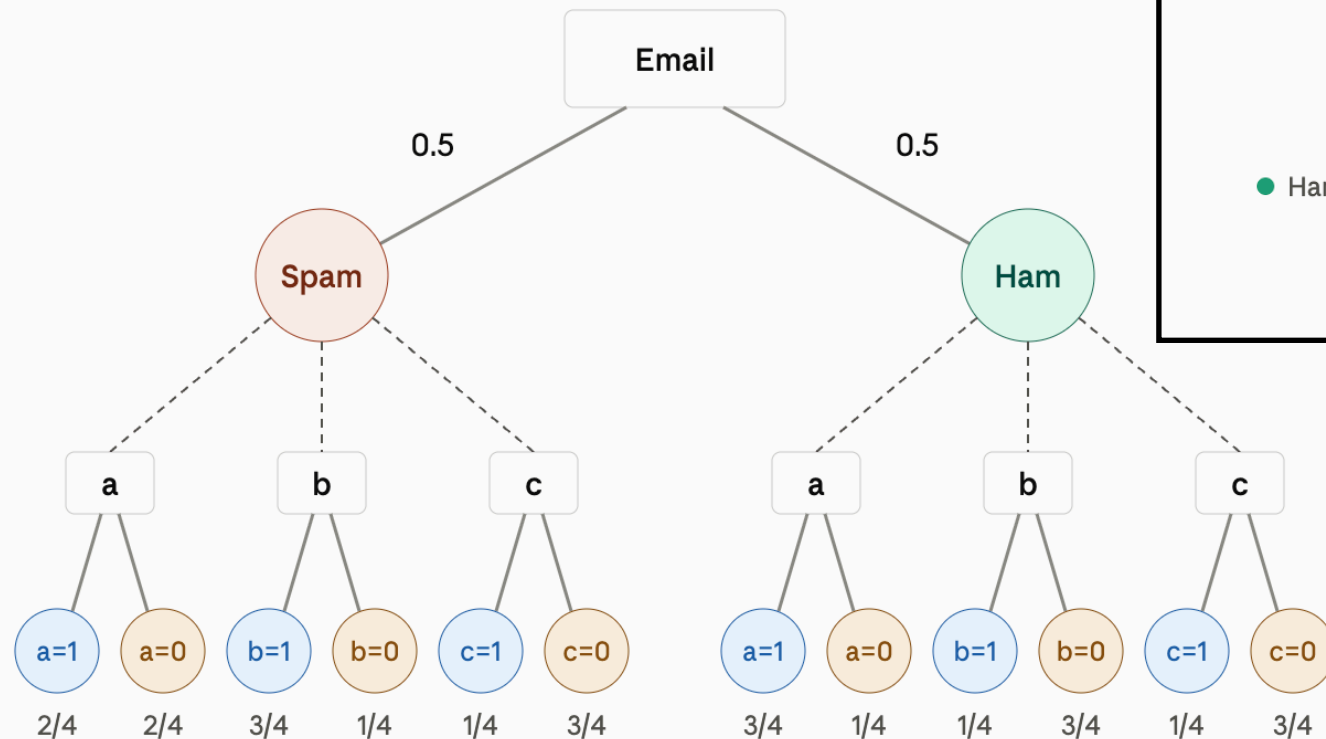
Q2c

Step2: Write down the probabilities



Q2c Analyze

Step2: Write down the probabilities



$\mathbf{x}_\star = (1, 1, 0)$ Q2c Analyze

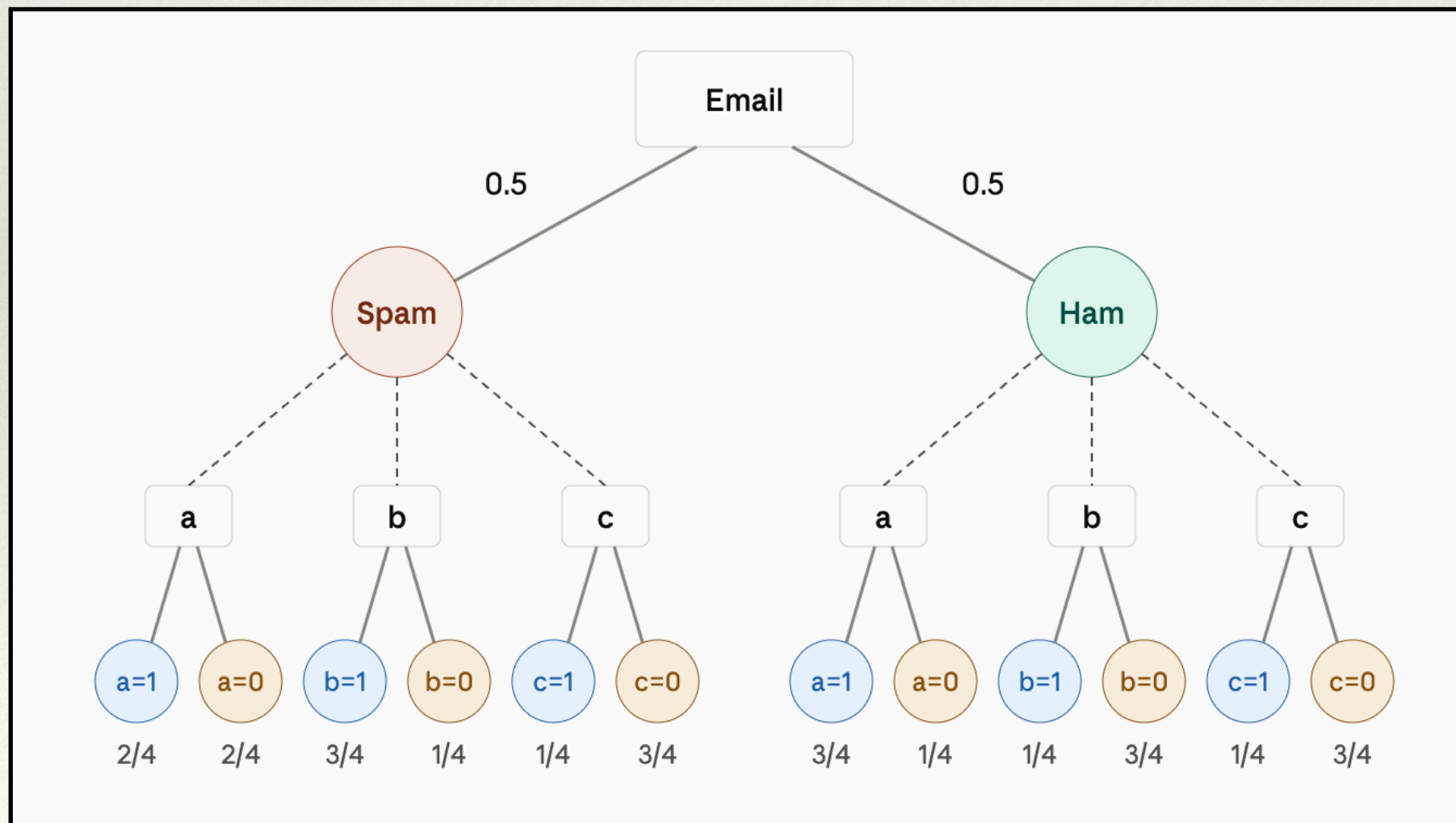
Step3: Apply Bayes' rule

Spam:

$$P(S \mid \mathbf{x}_\star) \propto P(a=1 \mid S) P(b=1 \mid S) P(c=0 \mid S) P(S)$$

Ham:

$$P(\sim S \mid \mathbf{x}_\star) \propto P(a=1 \mid \sim S) P(b=1 \mid \sim S) P(c=0 \mid \sim S) P(\sim S)$$



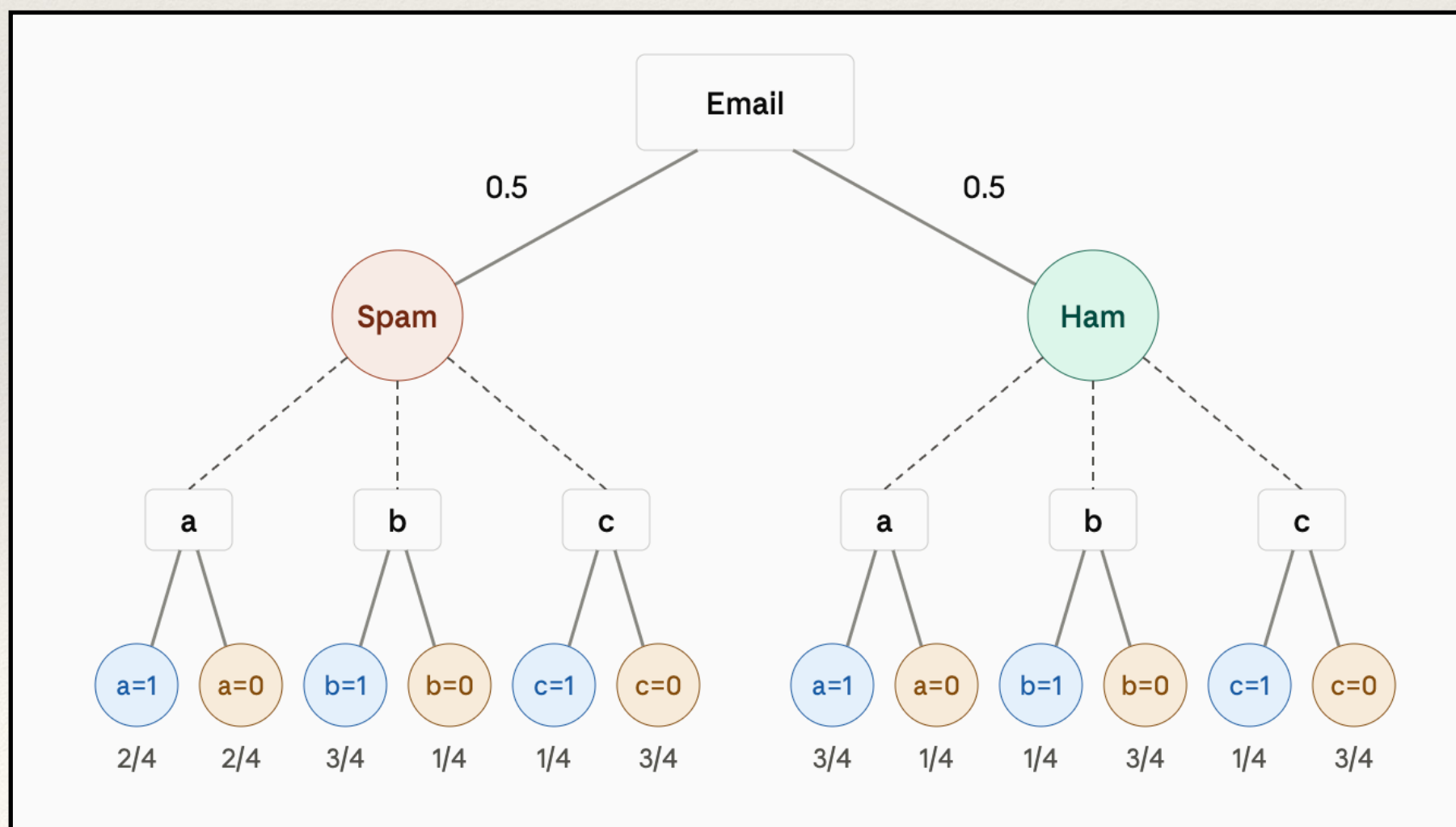
$$\mathbf{x}_\star = (1, 1, 0)$$

Q2c Calculation

Step3: Apply Bayes' rule

$$P(S \mid \mathbf{x}_\star) \propto P(a=1 \mid S) P(b=1 \mid S) P(c=0 \mid S) P(S) = \frac{2}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} = \frac{9}{64}$$

$$P(\sim S \mid \mathbf{x}_\star) \propto P(a=1 \mid \sim S) P(b=1 \mid \sim S) P(c=0 \mid \sim S) P(\sim S) = \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} = \frac{9}{128}$$

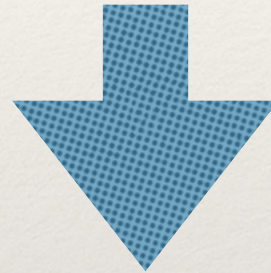


Q2c Calculation

$$P(S \mid \mathbf{x}_\star) \propto P(a=1 \mid S) P(b=1 \mid S) P(c=0 \mid S) P(S) = \frac{2}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} = \frac{9}{64}$$

$$P(\sim S \mid \mathbf{x}_\star) \propto P(a=1 \mid \sim S) P(b=1 \mid \sim S) P(c=0 \mid \sim S) P(\sim S) = \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} = \frac{9}{128}$$

Step4: Normalization



$$P(S \mid \mathbf{x}_\star) = \frac{\frac{9}{64}}{\frac{9}{64} + \frac{9}{128}} = \frac{2}{3} \approx 0.667$$

$$P(\sim S \mid \mathbf{x}_\star) = \frac{\frac{9}{128}}{\frac{9}{64} + \frac{9}{128}} = \frac{1}{3} \approx 0.333$$

$$P(S \mid \mathbf{x}_\star) > P(\sim S \mid \mathbf{x}_\star) \Rightarrow \text{predict spam}$$

Q2d Analyze

Without smoothing

$$p_j^k = \frac{N_{jk}}{N_k}$$

Can give probabilities of 0 or 1.

With Laplace smoothing

$$p_j^k = \frac{N_{jk} + 1}{N_k + V_j}$$

Avoids probabilities of 0 or 1.

- N_{jk} : documents in class k containing word j .
- N_k : total documents in class k .
- V_j : number of possible values of X_j (**2 for binary features**).

Q2d Analyze

- N_{jk} : documents in class k containing word j .
- N_k : total documents in class k .
- V_j : number of possible values of X_j (**2 for binary features**).

Without smoothing

$$p_j^k = \frac{N_{jk}}{N_k}$$

Can give probabilities of 0 or 1.

With Laplace smoothing

$$p_j^k = \frac{N_{jk} + 1}{N_k + V_j}$$

Avoids probabilities of 0 or 1.

$$p_a^+ = \frac{2}{4},$$
$$p_a^- = \frac{3}{4},$$

$$p_a^+ = \frac{2 + 1}{4 + 2} = \frac{3}{6},$$
$$p_a^- = \frac{3 + 1}{4 + 2} = \frac{4}{6},$$

Multinomial Naive Bayes

Q2e Analyze

e_1	b	d	e	b	b	d	e			
e_2	b	c	e	b	b	d	d	e	c	c
e_3	a	d	a	d	e	a	e	e		
e_4	b	a	d	b	e	d	a	b		
e_5	a	b	a	b	a	b	a	e	d	
e_6	a	c	a	c	a	c	a	e	d	
e_7	e	a	e	d	a	e	a			
e_8	d	e	d	e	d					

(e) Redo the previous analysis for the Multinomial Naive Bayes model without smoothing. Use the following test email: $e_\star = abbdebbcc$

Step1: Take a look at the dataset

Step2: Write down the probabilities

Just Calculate the sum

	x_{ia}	x_{ib}	x_{ic}
e_1	0	3	0
e_2	0	3	3
e_3	3	0	0
e_4	2	3	0
e_5	4	3	0
e_6	4	0	3
e_7	3	0	0
e_8	0	0	0

$$\theta_a^+ = \frac{5}{17}, \quad \theta_b^+ = \frac{9}{17}, \quad \theta_c^+ = \frac{3}{17},$$

$$\theta_a^- = \frac{11}{17}, \quad \theta_b^- = \frac{3}{17}, \quad \theta_c^- = \frac{3}{17}.$$

(1,4,2) $e_\star = abbdebbcc$

Q2e Analyze

$$\begin{aligned}\theta_a^+ &= \frac{5}{17}, & \theta_b^+ &= \frac{9}{17}, & \theta_c^+ &= \frac{3}{17}, \\ \theta_a^- &= \frac{11}{17}, & \theta_b^- &= \frac{3}{17}, & \theta_c^- &= \frac{3}{17}.\end{aligned}$$

Step3: Apply Bayes' rule

$e_\star = abbdebbcc$

$$x_a = 1, x_b = 4, x_c = 2$$

$$\begin{aligned}p(c_+ \mid \mathbf{x}_\star) &= \frac{p(c_+) p(\mathbf{x}_\star \mid c_+)}{p(\mathbf{x}_\star)} \\ &\propto p(c_+) p(\mathbf{x}_\star \mid c_+)\end{aligned}$$

Bayes Rule

$$\begin{aligned}&= p(c_+) \frac{n!}{x_a! x_b! x_c!} p(a \mid c_+)^{x_a} p(b \mid c_+)^{x_b} p(c \mid c_+)^{x_c} \\ &\propto p(c_+) p(a \mid c_+)^{x_a} p(b \mid c_+)^{x_b} p(c \mid c_+)^{x_c} \\ &= p(c_+) (\theta_a^+)^1 (\theta_b^+)^4 (\theta_c^+)^2\end{aligned}$$

multinomial
likelihood

Q2e Calculation

$$p(c_+ \mid \mathbf{x}_\star) \propto \frac{1}{2} \times \frac{5}{17} \left(\frac{9}{17}\right)^4 \left(\frac{3}{17}\right)^2 = \frac{295245}{2 \times 17^7} \approx 3.60 \times 10^{-4}$$

$$p(c_- \mid \mathbf{x}_\star) \propto \frac{1}{2} \times \frac{11}{17} \left(\frac{3}{17}\right)^4 \left(\frac{3}{17}\right)^2 = \frac{8019}{2 \times 17^7} \approx 9.77 \times 10^{-6}$$

Q2f

(f) Repeat the analysis for the smoothed Multinomial Naive Bayes model.

Model	Without smoothing	With Laplace smoothing
Bernoulli	$p_j^k = \frac{D_{jk}}{D_k}$	$p_j^k = \frac{D_{jk} + 1}{D_k + 2}$
Multinomial	$\theta_j^k = \frac{T_{jk}}{T_k}$	$\theta_j^k = \frac{T_{jk} + 1}{T_k + V }$

$|V|$: number of distinct words (**3 for a, b, c**).

Q2f

(f) Repeat the analysis for the smoothed Multinomial Naive Bayes model.

Model	Without smoothing	With Laplace smoothing
Bernoulli	$p_j^k = \frac{D_{jk}}{D_k}$	$p_j^k = \frac{D_{jk} + 1}{D_k + 2}$
Multinomial	$\theta_j^k = \frac{T_{jk}}{T_k}$	$\theta_j^k = \frac{T_{jk} + 1}{T_k + V }$

$|V|$: number of distinct words (**3** for a, b, c).

$$\theta_a^+ = \frac{5}{17},$$
$$\theta_a^- = \frac{11}{17},$$

$$\theta_a^+ = \frac{5 + 1}{17 + 3} = \frac{6}{20}, \quad \theta_b^+ = \frac{9 + 1}{17 + 3} = \frac{10}{20}, \quad \theta_c^+ = \frac{3 + 1}{17 + 3} = \frac{4}{20},$$
$$\theta_a^- = \frac{11 + 1}{17 + 3} = \frac{12}{20}, \quad \theta_b^- = \frac{3 + 1}{17 + 3} = \frac{4}{20}, \quad \theta_c^- = \frac{3 + 1}{17 + 3} = \frac{4}{20}.$$

Question 3

- (a) What is the role of the logistic sigmoid function $\sigma()$ in the logistic regression formulation? Why are we not able to simply use linear regression here? (a plot of $\sigma(z)$ may be helpful here).

Q3a Analyze

(a) What is the role of the logistic sigmoid function $\sigma()$ in the logistic regression formulation? Why are we not able to simply use linear regression here? (a plot of $\sigma(z)$ may be helpful here).



Thank you!